Alex Morehead

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Machine Learning • Deep Learning • Computational Biology • High-Performance Computing

Education

University of Missouri Columbia, Missouri
Ph.D. Computer Science Aug 2020 - Present

- *Dissertation Advisor*: Prof. Jianlin Cheng
- *Cumulative GPA*: 4.0/4.0
- *Relevant Coursework*: Computational Intelligence, Machine Learning/Pattern Recognition, Neural Networks, Computer Vision, NLP, Unsupervised Learning, Computational Systems Biology, Design & Analysis of Algorithms

Missouri Western State University St. Joseph, Missouri
B.S. Computer Science Aug 2016 - May 2020

- *Minor*: Mathematics
- *Cumulative GPA*: 4.0/4.0 with General Studies Honors

Professional Experience

Bioinformatics & Machine Learning Lab Columbia, Missouri
Graduate Research Assistant | Advisor: Jianlin Cheng Aug 2020 – Present

- Research and develop novel geometric deep learning algorithms applicable to computational biology
- Authored and publicly released the NSF-funded DeepInteract, a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts, along with the new Geometric Transformer
- Curated the NSF-funded DIPS-Plus, to date the largest feature-rich protein interface prediction dataset comprised of 42,000 protein complexes, and publicly released it alongside a dedicated manuscript

EnCircle Technologies Columbia, Missouri
Volunteer & Teaching Assistant | Manager: Jamie Bolyard Jun 2021 – Present

- Teach youth with autism spectrum disorder (ASD) technical skills for future employment and hobbies

Altec Information Services Team St. Joseph, Missouri (Virtual)
Software Development Intern | Managers: Dan White, Annie Martin Aug 2018 - Jan 2021

- Reduced miscommunication between service centers globally by engineering over 5 new Angular web applications
- Built more than 6 secure backend APIs with the Spring framework in an agile development setting
- Maintained the test environments of 3 in-house applications by writing LINQ queries and tuning up test databases

NSF Research Experience for Undergraduates in Data Science of Risk and Human Activity Indianapolis, Indiana
Undergraduate Research Assistant | Program Directors: George Mohler, Mohammad Al Hasan Jun 2019 - Aug 2019

- Authored an ensemble pipeline of 3 convolutional neural networks trained to detect gunshot sounds in the vicinity
- Deployed the pipeline to a cluster of Raspberry Pi 3 Model B+ microcomputers and evaluated its performance in real-world settings in collaboration with Indianapolis public safety officials
- Published as a manuscript and orally presented the gunshot sound detection project's results at the 2019 IEEE International Conference on Big Data

NSF Research Experience for Undergraduates in Synthetic Biology St. Joseph, Missouri
Undergraduate Research Assistant | Program Directors: Todd Eckdahl, Jeffrey Poet May 2018 - Aug 2018

- Investigated applications of combinatorial optimization to model lab experiments performed by synthetic biologists
- Produced Variant Sampler, a Java application for modeling the sample space of in vitro experiments
- Published results of the Variant Sampler project in the American Journal of Undergraduate Research

Center for Academic Support St. Joseph, Missouri
Computer Science Content Tutor | Manager: Karen Luke Feb 2017 - Aug 2018

- Tutored a total of 5 undergraduate students, 2 being from underrepresented groups
- Spent up to 1 hour with each student per week, reviewing important computer science topics such as data structures and object-oriented programming
- Designed and initiated a customized learning plan for each student's success

Publications

Accepted/Published

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22
M. Shoman, A. Aboah, **A. Morehead**, Y. Duan, A. Daud, Y. Adu-Gyamfi — New Orleans, LA

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22
A. Morehead, C. Chen, J. Cheng — (Virtual)

High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function — IEEE SuperComputing '21, St. Louis, MO
M. Gao, P. L. Andersen, **A. Morehead**, S. Mahmud, C. Chen, X. Chen, N. Giri, R. Roy, F. Quadir, T. C. Effler, R. Prout, S. Abraham, W. Elwasif, J. Skolnick, J. Cheng, A. Sedova

Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo — AJUR '20, Journal
O. Koucky, J. Wagner, S. Aguilera, B. Bashaw, Q. Chen, A. Eckdahl, E. Edman, P. Gomez, N. Hanlan, N. Kempf, D. Mattoon, S. McKlin, C. Mazariegos, **A. Morehead**, S. Q. Ong, A. Peterson, M. Rojas, K. Roland, K. Schildknecht, H. Seligmann, K. Slater, A. Tauchen, R. Tittor, T. Travieso, D. Urban, C. Willis, J. Zhou, N. L. Snyder, L. J. Heyer, J. L. Poet, T. T. Eckdahl, A. M. Campbell

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19
A. Morehead, L. Ogden, G. Magee, R. Hosler, B. White, G. Mohler — Los Angeles, CA

Under Review

DRLComplex: Reconstruction of protein quaternary structures using deep reinforcement learning — arXiv '22
E. Soltanikazemi, R. S. Roy, F. Quadir, N. Giri, **A. Morehead**, J. Cheng — Under Review

DProQ: A Gated-Graph Transformer for Protein Complex Structure Assessment — arXiv '22
X. Chen, **A. Morehead**, J. Liu, J. Cheng — Under Review

EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures — arXiv '22
A. Morehead, X. Chen, T. Wu, J. Liu, J. Cheng — Under Review

3D-Equivariant Graph Neural Networks for Protein Model Quality Assessment — bioRxiv '22
C. Chen, X. Chen, **A. Morehead**, T. Wu, J. Cheng — Under Review

Preprint

Semi-Supervised Graph Learning Meets Dimensionality Reduction — arXiv '22
A. Morehead, W. Chantapakul, J. Cheng — Preprint Archive

DIPS-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction — arXiv '21, Preprint Archive
A. Morehead, C. Chen, A. Sedova, J. Cheng

Presentations

Geometric Transformers for Protein Interface Contact Prediction — Shandong University '22
A. Morehead — Qingdao, Shandong Province, China (Virtual)

High-Performance Toolbox for Deep Learning of Protein Structure and Function — IEEE SuperComputing '21
A. Morehead, A. Sedova, M. Gao — St. Louis, Missouri (Virtual)

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19
A. Morehead, L. Ogden, G. Magee — Los Angeles, California

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IUPUI Student Summer Poster Symposium
A. Morehead, L. Ogden, G. Magee — Indianapolis, Indiana

Variant Sampling in vitro with a Scheduling Twist — Alpha Chi National Convention '19
A. Morehead — Cleveland, Ohio

Variant Sampling in vitro with a Scheduling Twist MWSU PORTAL Summer Research Showcase '18
A. Morehead St. Joseph, Missouri

Predicting Game Genres by Analyzing Code Structure CSCC Central Plains Conference '18
S. Frazier, **A. Morehead**, S. Prine Maryville, Missouri

Predicting Game Genres by Analyzing Code Structure MWSU Multidisciplinary Research Day '18
S. Frazier, **A. Morehead**, S. Prine St. Joseph, Missouri

Leadership Activities

ICLR Volunteer Apr 2022
NeurIPS Reviewer Sep 2021 – Present
Upsilon Pi Epsilon (ΥΠΕ) | University of Missouri Chapter Aug 2020 - Present
Alpha Chi (AX) | Missouri Western State University Chapter Mar 2018 – May 2020
Kappa Mu Epsilon (KME) | Missouri Western State University Chapter Mar 2018 – May 2020

Awards & Grants

CVPR Travel Grant *May 2022*
XSEDE Startup Hardware Grant *Dec 2021*
Dean's Engineering Excellence Fellowship *Aug 2020*
James W. and Joan M. O'Neill Graduate Fellowship in Engineering *Aug 2020*
MWSU Outstanding Graduating Computer Science Student Award *May 2020*
MWSU President's Honor Roll *May 2020*
Floyd Tesmer/Strayer University Prize in Computer Science and Engineering *Apr 2019*
Alpha Chi Region IV Scholarship *Apr 2019*
Grand Midwest Asynchronous Programming Contest 3rd Place Prize *Apr 2017*
East Side Lions Club Scholarship *May 2016*

Projects

BioPython (3.1k Stars) | GitHub: https://github.com/biopython/biopython Apr 2022 – Present
- Created functionality to remove large feature generation temporary files after all processing completes
- Collaborated with senior project developers to ensure proposed solution matched existing code standards

Graphein (300+ Stars) | GitHub: https://github.com/a-r-j/graphein Apr 2022 – Present
- Added workflow to catch illusive OS errors when executing external programs
- Worked with senior project developer to ensure workflow matched existing code standards

EGNN-PyTorch (150+ Stars) | GitHub: https://github.com/lucidrains/egnn-pytorch Apr 2021 – Present
- Noticed the omission of an important geometric unit test for this neural network architecture
- Developed and validated the code to test for node-wise permutation equivariance within the network

DeepInteract (20+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DeepInteract Oct 2021 – Present
- Designed and developed a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Maintain project by addressing bug fixes, feature requests, and future planning

DIPS-Plus (15+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DIPS-Plus June 2021 – Present
- Curated (to date) the largest molecular dataset for machine learning of protein interface contacts
- Work to ensure ease of use and adoption by responding to feature requests and open issues

Technical Skills

Programming: Python 3 • Java • Angular 2+ • C/C++ • HTML/CSS • SQL • C# • R • MATLAB

Tools/Frameworks: PyTorch • TensorFlow • NumPy • Pandas • Spring Boot • NodeJS • Docker • AWS • Google Cloud

Methodologies: RESTFUL APIs • Version Control {Git} • Agile Development

News & Media Outreach

[1] MU Engineering. (2020, October 27). *Alex Morehead*. University of Missouri College of Engineering. https://engineering.missouri.edu/2020/10/alex-morehead/.

[2] MU Engineering. (2021, October 18). *New protein prediction tool could accelerate biological discoveries*. University of Missouri College of Engineering. https://engineering.missouri.edu/2021/10/new-protein-prediction-tool-could-accelerate-biological-discoveries/.

[3] ORNL. (2021, January 10). *Summit Powers Novel Protein Function Prediction Work*. Oak Ridge National Laboratory. https://www.ornl.gov/news/scientists-use-summit-supercomputer-deep-learning-predict-protein-functions-genome-scale/.

[4] HPCwire. (2021, January 13). *Summit Powers Novel Protein Function Prediction Work*. HPCwire. https://www.hpcwire.com/2022/01/13/summit-powers-novel-protein-function-prediction-work/.

[5] MarkTechPost. (2022, January 18). *Researchers Introduce High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function*. MarkTechPost Tech News – AI Paper Summary. https://www.marktechpost.com/2022/01/18/researchers-introduce-high-performance-deep-learning-toolbox-for-genome-scale-prediction-of-protein-structure-and-function/.